

Lesson 06

Division Algorithm - I

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The steps to study this lesson

In the last lecturer, can you remember, we learned about Division Algorithm, the proof and the applications of it. In this written lesson, I am exposit it more for your convenience. Please thoroughly study this lesson by following the steps given below.

- Recall the famous quote from George Polya “Mathematics is not a spectator’s game.” If you doing mathematics always keep this in mind.
- Recall your past experiences on the “division algorithm”(actually you didn’t use this word at that time) in your primary and secondary school.
- Carefully read the statement on theorem 6.2.1. Relate the statement with your past experiences with the “division algorithm”. Note the difference in this statement, here the dividend need not necessarily be positive. But in your school days the always dividend was a positive integer.
- Recall the Well Ordering Principle. We are going to prove the Division Algorithm, by applying the Well Ordering Principle. To apply the Well Ordering Principle for some sub set of non negative integers that set should be non empty. Please carefully complete the Activity 1, which is useful for you to understand the proof of Division Algorithm.

- Carefully study the proof of Division Algorithm. Note that, the proof consists of two parts. The existence part and the uniqueness part.
- Carefully study how to work out Example 6.3.1, Example 6.3.2 and Example 6.3.3. First you try to complete these examples by your own. After that you can study answers written in the note.
- Complete the Activity 02.

6.1 Introduction

Now we are ready to establish a very familiar experience from your school days, the division algorithm. Division Algorithm is nothing, but if you divide some integer with another positive integer (say b) the possible remainders are $0, 1, 2, \dots (b - 1)$. Also this remainder as well as the quotient is unique. For someone, that can be seen as trivial fact. But in rigorous mathematics, it is always asking proofs. Here we present a rigorous proof in abstract setup. Again the heart of the proof is Well Ordering Principle. The proof consists of two major parts, existence and the uniqueness. Before studying the theorem, do the following activity.

Activity 01

Let $a \in \mathbb{Z}$, $b \in \mathbb{N}$. Prove the set $S = \{a - xb : x \in \mathbb{Z}, a - xb \geq 0\}$ is non empty.

{Hint:-Consider two cases (i) $a > 0$ and (ii) $a \leq 0$ }

6.2 Division Algorithm

6.2.1 Theorem -Division Algorithm

Given integers a and b , with $b > 0$ there exist unique integers q and r satisfying $a = qb + r$ with $0 \leq r < b$.

The integers q and r are called, respectively, the quotient and remainder in the division of a by b .

Proof:

Consider the set $S = \{a - xb : x \in \mathbb{Z}, a - xb \geq 0\}$.

To apply the well ordering principle let us first prove that this set is a non empty set of non negative integers.

Since a , x , and b are integers, it is clear that $a - xb$ is an integer. Now to show this set is non empty, we must show that there is an integer x , such that $a - xb \geq 0$. Recall that b is a positive integer and hence $b \geq 1$.

If $a > 0$ choose $x = -a$

Then it is clear that $a - xb = a + ab \geq 0$.

If $a \leq 0$ choose $x = a$.

Then $a - xb = a - ab = a(1 - b) = -a(b - 1)$.

But $b - 1 \geq 0$ and $-a \geq 0$.

Hence $a - xb \geq 0$

Therefore, the set S is non empty.

Now by well ordering principle the set S contains a least element. Let it be r .

By definition of S , there exist an integer q such that $r = a - bq$.

Suppose $r \geq b$,

Then $a - (q + 1)b = (a - qb) - b = r - b \geq 0$

This implies that the integer $a - (q + 1)b$ in the set S .

But $-(q + 1)b = r - b < r$, leading to a contradiction with the fact that r is the least element of S . Hence $r < b$.

Now let's prove the uniqueness of q and r .

Suppose a has two representations of the form say, $a = qb + r = q'b + r'$ with $0 \leq r < b$ and $0 \leq r' < b$.

Now since, $qb + r = q'b + r'$,

$$|r - r'| = |b(q' - q)| = b|q - q'|.$$

Since $0 \leq r < b$ and $0 \leq r' < b$ we have that $|r - r'| < b$.

Thus $b|(q' - q)| < b$, which yields $0 \leq |(q' - q)| < 1$.

Since $|(q' - q)|$ is none negative integer, the only possible way to satisfies this inequality is $q = q'$. This result initially gives $r = r'$. ■

6.3 Applications of Division Algorithm

Though the Division Algorithm is very simple concept, there are various applications on it. The following are some examples.

6.3.1 Example

Using division algorithm prove that for $n \in \mathbb{Z}$, $3|n(2n^2 + 7)$

Answer:-

Let n be an arbitrary integer. Then by Division Algorithm $n = 3q + r$, $0 \leq r < 3$.

Hence $r = 0, 1$ or 2 .

when $r = 0$, then $3|n$ and hence $3|n(2n^2 + 7)$

when $r = 1$, then $n = 3q + 1$. Hence $2n^2 + 7 = 2(3q + 1)^2 + 7 = 18q^2 + 12q + 9$

But $18q^2 + 12q + 9 = 3(6q^2 + 4q + 3)$. Therefore $3|n(2n^2 + 7)$.

when $r = 2$, then $n = 3q + 2$. Hence $2n^2 + 7 = 2(3q + 2)^2 + 7 = 18q^2 + 24q + 15$

But $18q^2 + 24q + 15 = 3(6q^2 + 8q + 5)$. Which implies $3|n(2n^2 + 7)$.

Hence for every $n \in \mathbb{Z}$, $3|n(2n^2 + 7)$.

6.3.2 Example

Prove that, product of three consecutive integers is divisible by 6.

Answer:

Let n be an arbitrary integer. Then by Division Algorithm $n = 6q + r$, $0 \leq r < 6$.

Hence $r = 0, 1, 2, 3, 4$, or 5 .

If $r = 0$, then $6|n$ and hence $6|n(n + 1)(n + 2)$.

If $r = 1$, then $n = 6q + 1$, $n + 1 = 6q + 2 = 2(3q + 1)$, $n + 2 = 6q + 3 = 3(2q + 1)$ and hence $2|(n + 1)$ and $3|(n + 2)$. This implies $6|n(n + 1)(n + 2)$.

If $r = 2$, then $n = 6q + 2 = 2(3q + 1)$, $n + 1 = 6q + 3 = 3(2q + 1)$ and hence $2|n$ and $3|(n + 1)$. This implies $6|n(n + 1)(n + 2)$.

If $r = 3$, then $n = 6q + 3 = 3(2q + 1)$, $n + 1 = 6q + 4 = 2(3q + 2)$ and hence $3|n$ and $2|(n + 1)$. This implies $6|n(n + 1)(n + 2)$.

If $r = 4$, then $n + 2 = 6q + 6 = 6(q + 1)$ and hence $6|n(n + 1)(n + 2)$.

If $r = 5$, then $n + 1 = 6q + 6 = 6(q + 1)$ and hence $6|n(n + 1)(n + 2)$.

Hence the result.

Also one can write the proof in following simple way. Though the following proof is not wrong, a major explanation is missing. You will learn about it more in a later session.

“From any two consecutive integers exactly one must be even. Hence from any three consecutive integers at least one must be even.

From any three consecutive integers at least one must be divided by three

Hence product of three consecutive integers is divisible by 6”.

Activity 02

- (i) Prove that from any four consecutive integers at least one must be divided by 4.
- (ii) Prove that for any integer a , one of the integers a , $a + 2$, $a + 4$ is divisible by 3.
- (iii) Prove that product of any three consecutive even numbers is divisible by 48.

Following examples are diverse applications of division algorithm. For some problems you can apply the division algorithm directly. But for some problems you must be more careful and for efficiently solve the problem division algorithm must be use wisely.

6.3.3 Examples

Use the division algorithm to establish that;

- (i) The square of any integer is either of the form $3k$ or $3k + 1$
- (ii) The fourth power of any integer is either of the form $5k$ or $5k + 1$
- (iii) An integer of the form $4k + 2$ never a perfect square.

- (iv) An integer of the form $4k + 3$, cannot be written as sum of two perfect squares.
(v) For any odd integer, $n^4 + 4n + 11$ is divisible by 16.

Answers:-

(i) Let n be any integer. Then by division algorithm n is of the form $3k$, $3k + 1$ or $3k + 2$. For each case, n^2 is equal to $9k^2$, $(9k^2 + 6k + 1)$ and $(9k^2 + 12k + 4)$ respectively.

Since $9k^2 = 3 \cdot 3k^2$ and $9k^2 + 6k + 1 = 3(3k^2 + 2k) + 1$, those are of the form $3k$ and $3k + 1$ respectively. Also $9k^2 + 12k + 4 = 3(3k^2 + 4k + 1) + 1$. Hence square of the form $3k + 2$ is of the form $3k + 1$. Therefore the square of any integer is either of the form $3k$ or $3k + 1$.

(ii) Let n be any integer. Then by division algorithm n is of the form $5k$, $5k + 1$, $5k + 2$, $5k + 3$ or $5k + 4$. Observe that,

$$(5k + r)^4 = (5k)^4 + {}^4C_1(5k)^3r + {}^4C_2(5k)^2r^2 + {}^4C_3(5k)r^3 + r^4$$

Observe that for $r = 0, 1, 2, 3, 4$, the first four terms of the right hand side of the above identity is divided by 5. Possible values for r^4 is 0, 1, 16, 81, 256. But $16=5 \cdot 3+1$, $81=5 \cdot 16+1$, $256=5 \cdot 51+1$.

This shows that $(5k + r)^4$ is of the form $5k$ or $5k + 1$ for $r = 0, 1, 2, 3, 4$.

Hence the fourth power of any integer is either of the form $5k$ or $5k + 1$.

(iii) Let n be any integer. By division algorithm n is of the form $4k$, $4k + 1$, $4k + 2$ or $4k + 3$.

Now $(4k)^2 = 4 \cdot 4k^2$, $(4k + 1)^2 = 4(4k^2 + 2k) + 1$, $(4k + 2)^2 = 4(4k^2 + 4k + 1)$ and $(4k + 3)^2 = 4(4k^2 + 6k + 2) + 1$. Hence square of any integer is either of the form $4k$ or $4k + 1$. Therefore an integer of the form $4k + 2$ is never a perfect square.

(iv) Let an integer n be a sum of two square numbers. Hence $n = a^2 + b^2$ for some integers a and b . By (iii) a^2 is of the form $4k$ or $4k + 1$ as well as b^2 is of the form $4m$ or $4m + 1$.

But up to symmetry, by considering all the possibilities, $4k + 4m = 4(k + m)$,

$(4k + 1) + 4m = 4(k + m) + 1$, and $(4k + 1) + (4m + 1) = 4(k + m) + 2$. Hence the sum of two square numbers are of the form $4k$, $4k + 1$ or $4k + 2$. Hence an integer of the form $4k + 3$, cannot be written as sum of two perfect squares.

(v) At first looking, one may think, the following method is working.

By division algorithm $n = 16q + r$, where $0 \leq r < 16$. But since n is odd, the only values that r can be taken is 1, 3, 5, 7, 11, 13, 15. Now in each case substituting this in the equation $n^4 + 4n + 11$, one can show that each case this is a multiple of 16. Of course that will work. But that is very long method.

Let's see how to do it in a short method. Since n is even, $n = 2k + 1$ for some $k \in \mathbb{Z}$. Now
$$n^4 + 4n + 11 = (2k + 1)^4 + 4(2k + 1) + 11 = 16(k^4 + k + 1).$$

Hence $n^4 + 4n + 11$ is divisible by 16.